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SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

K POORNA CHANDU 192312585RC

CO2 - AT2 - Scenario Based Questions

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QUESTIONS

- **Q1** Scenario Based - 1. Electric Vehicle Battery Manufacturing System — 25 marks
- **Q2** Scenario Based - 2. Smart Grid Power Distribution System — 25 marks
- **Q3** Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026 — 25 marks
- **Q4** Scenario Based - Airport Passenger Security Screening System — 25 marks

4/4 answered

Q1 **Scenario Based - 1. Electric Vehicle Battery Manufacturing System** 25 marks

An electric vehicle battery manufacturing company produced batteries from January 2026 to December 2026. Each month, the target production is 5,000 batteries. If production exceeds the target, the company receives a 10% performance incentive. If production is below 80% of the target, the production unit is marked for inspection.

A) Develop a C program using arithmetic operators to calculate the annual production, average monthly production, and production efficiency. Use decision-making statements to determine the incentive and inspection status.

B) Use looping constructs to process production details for all 12 months. Use continue to skip months with invalid production values and break if the plant shuts down due to equipment failure.

Your Code

```
#include <stdio.h>
int main(){
    int load[15], maintenance[15];
    int i;
    float percentage, available;
    for(i = 0; i < 15; i++)
    {
        scanf("%d", &load[i]);
        scanf("%d", &maintenance[i]);
    }
    for(i = 0; i < 15; i++)
```

Input

```
250
0
300
0
```

Output

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Q4 Scenario Based - Airport Passenger Security Screening System
25 marks

4/4 answered

Q2 Scenario Based - 2. Smart Grid Power Distribution System

25 marks

A power distribution company monitors 15 substations every hour. Each substation has a maximum capacity of 500 MW.

Business Rules:

Load ≤ 60% → Normal

61–85% → High Load

Above 85% → Critical

Above 500 MW → Grid Failure

A) Develop a C program using arithmetic operators to calculate the load percentage and available capacity. Use decision-making statements to classify each substation.

B) Use looping constructs to monitor all substations. Use continue to skip substations under maintenance and break if grid failure occurs.

Your Code

```
#include <stdio.h>
int main() {
    int maxCapacity = 500;
    int subLoad[15];
    int maintenance[15];
    for (int i = 0; i < 15; i++) {
        printf("Substation %d load (MW): ", i + 1);
        scanf("%d", &subLoad[i]);
```

Input

```
450 0
505 0
300 1
200 0
```

Output

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Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026
25 marks

Q4
Scenario Based - Airport Passenger Security Screening System
25 marks

4/4 answered

Q3 **Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026** 25 marks

Vaccines are distributed to 25 hospitals. Each hospital must receive at least 8,000 doses.

Distribution Status:

≥10,000 doses → Sufficient

8,000–9,999 → Minimum Requirement

Below 8,000 → Emergency Supply Required

A) Develop a C program using arithmetic operators to calculate remaining vaccine stock after distribution. Use decision-making statements to classify each hospital.

B) Use looping constructs to process all 25 hospitals. Use continue for closed hospitals and break if stock becomes zero.

Your Code

```
#include <stdio.h>
int main(){
    int doses[25], closed[25];
    int stock = 250000;
    int i;
    for(i = 0; i < 25; i++)
    {
        scanf("%d", &doses[i]);
        scanf("%d", &closed[i]);
```

Input

```
12000
0
9000
0
```

Output

35°C
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Search

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IN

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03-09-2026

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Scenario Based - Airport Passenger Security Screening System
25 marks

4/4 answered

Q4 Scenario Based - Airport Passenger Security Screening System 25 marks

An international airport processes 1,500 passengers daily.

Security Rules:

Age below 18 → Minor Screening

Age 18–59 → Normal Screening

Age 60 and above → Senior Citizen Screening

Carrying baggage above 30 kg → Additional Inspection

Carrying prohibited items → Entry Denied

A) Develop a C program using arithmetic operators to calculate total baggage weight and screening statistics. Use decision-making statements to determine passenger screening status.

B) Use looping constructs to process passengers. Use continue to skip cancelled tickets and break during emergency evacuation.

Your Code

```
#include <stdio.h>
int main(){
    int age[10], weight[10], prohibited[10], cancelled[10];
    int i, totalWeight = 0;
    for(i = 0; i < 10; i++)
    {
        scanf("%d", &age[i]);
```

Input

```
15
20
0
0
```

Output